

Where Next? Predicting the Next Unaffordable U.S. City

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Abstract

Housing affordability collapse hits mid-sized U.S. metros with no warning - this project builds an early-warning model trained on Tampa, Austin, and Boise to flag which current affordable cities are next. Using five years of housing, income, and population data joined via CBSA metro codes, the model surfaces the leading indicators that preceded collapse in these three metros and applies them to score at-risk today.



Introduction & Motivation

Problem & question:

Which socioeconomic and housing market indicators best predict the onset of affordability collapse in mid-sized U.S. metros, and can those indicators identify at-risk cities before displacements begins?

Impact & stakeholders:

City housing departments, regional lender, and real estate investors need lead time to act; working- and middle-class residents in Tampa, Austin, and Boise experienced this collapse without warning, and the next wave of affected cities shouldn't have to.



Background & Related Work

Zillow Research and the FHFA publish current-state affordability indices, but neither is built to predict onset - they describe where a metro stand today, not where it's headed. The Urban Institute's housing cost burden research frames displacement risk in terms of income and rent ratios but stops short of a forward-look, metro-level early-warning score. This project differs by treating Tampa, Austin, and Boise as labeled "collapsed" trajectories and asking which pre-collapse signals generalize to other metros.



Data, Engineering & Ethics

Sources & scale: Zillow ZHVI (median home values), FHFA HPI (quarterly price appreciation), Census ACS (income, cost burden, demographics), and Census Population Estimates - all joined through CBSA metro codes

Pipeline: ingest (R and Python) → clean (R and Python) → transform (R)→ engineered features (Python)→ store (GitHub).

Ethics & responsible use: Demographic variables are excluded from the predictive model and used only in post-hoc equity checks, since a false "at-risk" flag could deter needed investment and a missed flag could mean residents get no warning at all.

[Pipeline sketch: boxes & arrows]



Methods & Evaluation

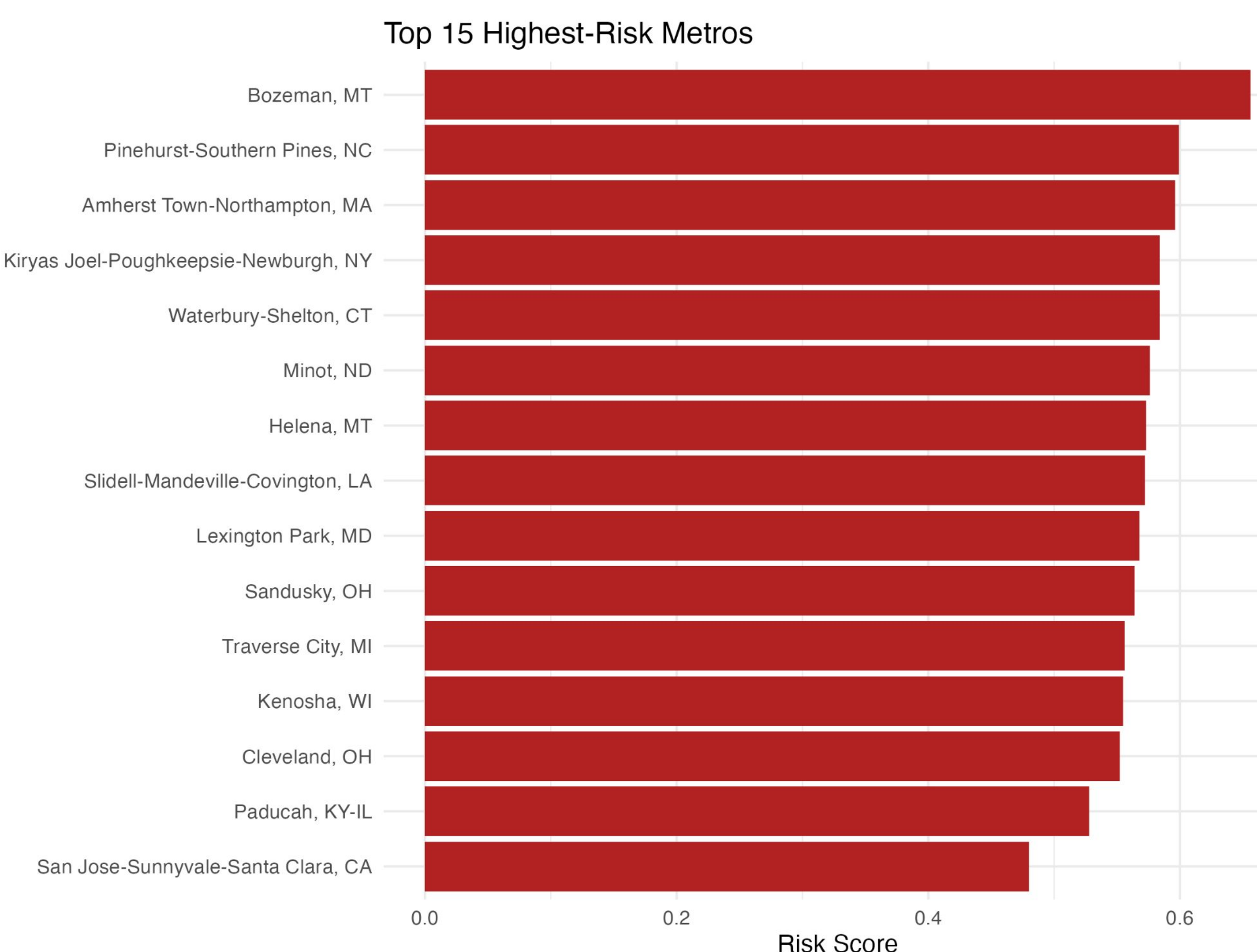
Approach: An XGBoost classifier trained on engineered features - price-to-income ratio trend, affordability momentum score, housing supply pressure, mortgage burden % of income, and population velocity. Then using Tampa, Austin, Boise as labeled collapse trajectory, then score against still-affordable comparison metros.

Validation: Backtesting holds out each training city's final 2-3 pre-collapse years to test whether the model would have flagged it early using only earlier data; R-based Granger causality test confirm indicators lead rather than lag price surgages, and along with multiple cross-validation testing and tuning.



Results & Visualizations

The model's top-15 riskiest metros span both small boomtowns like Bozeman, MT and already pricey hubs like San Jose, CA - showing that affordability collapse is driven by the rate of deterioration, not current price level or metro size.



Discussion, Limitations & Conclusions

Discussion: A high risk score means a metro's current trajectory resembles Tampa, Austin, or Boise 3-5 years before their collapse - not that collapse is guaranteed.

Limitations & future work: The three-city training set is small and geographically skewed towards Sun Belt/mountain-west boomtowns, so results may not generalize to older Rust Belt or Northeast metros; ACS data lags by roughly a year, meaning recent shifts may not appear in the model yet; and despite Granger tests, correlation-driven false signal remain possible.

Conclusions: The model identifies Bozeman, MT, Pinehurst-Southern Pines, NC, and Amherst Town-Northampton, MA as the metros most closely tracking the pre-collapse trajectory seen in Tampa, Austin, and Boise, suggesting regional housing authorities and mortgage lenders in these markets should prioritize supply-side interventions and underwriting caution now, before displacement pressure fully materializes.



Reproducibility & References

Repository: [<https://github.com/your-org/your-project>] · Key dependencies: [Python 3.x (pandas, scikit-learn, xgboost, shap), R (dplyr, lme4)]

References: Zillow Research ZHVI methodology; FHFA House Price Index documentation, Urban Institute housing cost-burden reports

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